

HAT-P-20b

R-1018

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_251220 · created 2026-07-30T04:12:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461029.9607 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.014
Transit depth (Rp/Rs)^2	1.83 +/- 0.37 [%]
Orbital Inclination (inc)	86.26 +/- 0.53
Airmass coefficient 1 (a1)	1.004 +/- 0.0091
Airmass coefficient 2 (a2)	-0.003 +/- 0.01
Scatter in the residuals of the lightcurve fit is	0.9 %
Best Comparison Star	#2 - [448, 373]
Optimal Method	PSF photometry
Transit Duration (day)	0.0707 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.135 ± 0.014 vs 0.1284 (deviation 0.45σ)
Duration fit vs published	0.0707 d vs 0.0775 d (deviation 0.82σ)
Mid-time O–C	1.0 min
Depth SNR	9.2
Residual scatter	0.9 %

Light curve

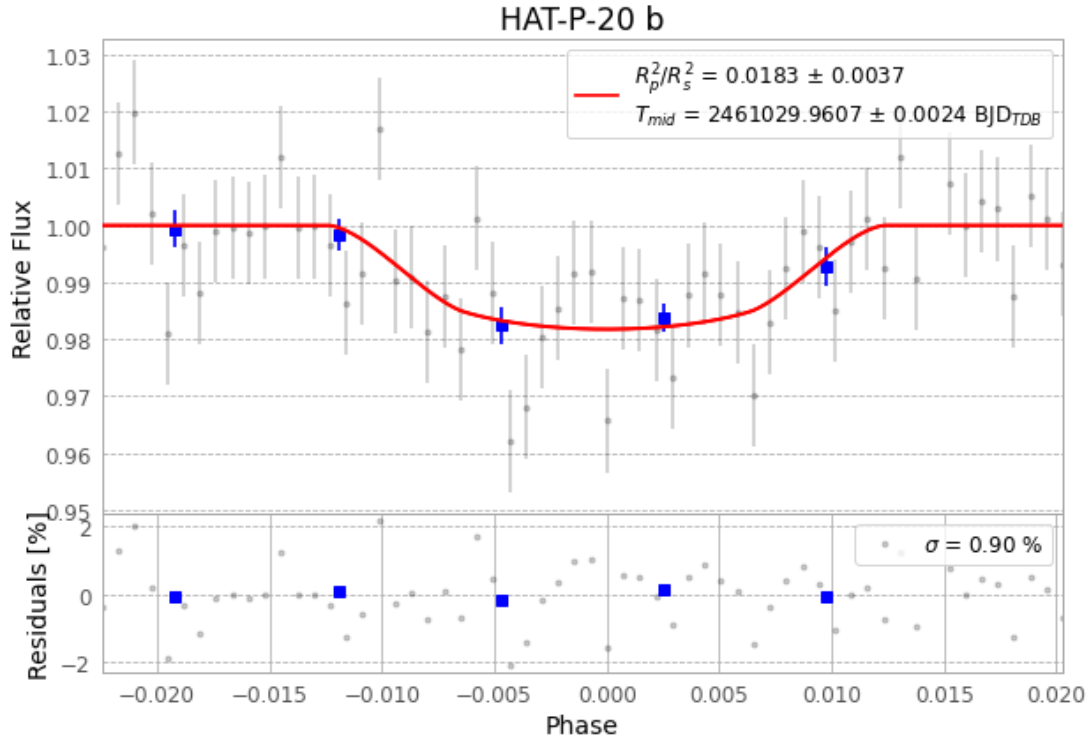


Figure 1 — FinalLightCurve_HAT-P-20 b_2025-12-20.png (SHA-256 e700cf3c10750325...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 321], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 acd92a47b917b19c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

60 FITS frames (39 MB), HATP-20251220092115.FITS → HATP-20251220121815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-251220.log (782bca36a045...), FinalLightCurve_HAT-P-20 b_2025-12-20.png (e700cf3c1075...), FinalLightCurve_HAT-P-20 b_2025-12-20.csv (356e31df0887...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 04:12Z.